

Mika Be

APPLIED AI | SOFTWARE ENGINEERING

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EDUCATION

Rutgers University, New Brunswick | B.S. Electrical and Computer Engineering, CS minor Expected May 2028

GPA 3.838/4.000 | Dean's List, all four completed semesters

Honors 2nd Place Best Full Paper, IEEE ISEC 2026 | IEEE-HKN honor society | Bergen County Academies CS Award (1 of 3)

Coursework Data Structures; Digital Logic Design; Linear Systems and Signals (in progress); Computer Architecture (in progress)

RESEARCH

Kuiper TTS, Rutgers ECE | Undergraduate Researcher and Team Lead (PI: Prof. Dov Kruger) Oct 2025-Present

Python | *PyTorch Lightning* | *Piper/VITS* | *ONNX*

- Co-authored an IEEE ISEC 2026 paper (3rd of 11 authors) on turning scientific papers into audio for blind and low-vision readers
- Led the 18-person research team: helped recruit members, assigned codebase tasks, ran the text-to-speech research
- Wrote narration rules for spoken math, e.g. “d q over d t” becomes “the derivative of q with respect to t”
- Rebuilt the training stack as a one-command GPU pipeline with ONNX export; fine-tuned 5 voices on 200 self-recorded clips
- Wrote voice-evaluation scripts (MCD, F0 RMSE, PESQ, STOI) and the guide the team uses to train new voices

EXPERIENCE

this.is | Lead Software Developer (Internship) Jul 2025-Aug 2026

React | *TypeScript* | *Firebase* | *Cloud Functions*

- Built a place-discovery web app as primary engineer (search, profiles, saved lists, sharing); mentored and managed 2 interns
- Prototyped embedding-based search; wrote a load-test harness for p50/p95/p99 latency and per-query cost at 300 users
- Designed and built v2, a private group app that ranks members' saved places into shared plan suggestions (in preview)

Schieber Research | Software Engineering Intern Sep 2023-Jun 2024

- Built and documented a React/Vite research prototype backed by Appwrite, turning study requirements into working flows

Starbucks | Barista (part-time, alongside full-time coursework) Jul 2023-Jun 2025

PROJECTS

Room of Days | Creator and Solo Developer, life-RPG habit app Jun 2026-Present

Flutter | *Dart* | *Swift (WidgetKit)* | *Firebase* | *Codemagic CI*

- Published on the App Store (Aug 2026) and the web (roomofdays.com); 10+ active users, 1,200+ automated tests
- Wrote a native Swift (WidgetKit) Home Screen widget alongside the Flutter app, plus .ics class-schedule import with reminders
- Designed local-first storage with versioned saves; cloud sync never overwrites an unreadable or newer-format save

Room Notes | Creator and Solo Developer, cross-device lecture notebook Aug 2026-Present

Flutter | *Dart* | *Objective-C++* | *Drift/SQLite* | *Firestore*

- Patched Flutter's iOS engine to deliver the between-frame Apple Pencil samples (coalesced touches) that stock Flutter ignores; compiled the custom engine in CI and shipped it to TestFlight
- Built local-first Drift/SQLite storage with a transactional outbox so edits survive offline use and app restarts

Knowsy Breadboard Lab, Rutgers ECE | Technical Lead, 3-person team May 2026-Present

JavaScript | *Verilog*

- Built a browser breadboard for Rutgers Digital Logic Design and Principles of EE II labs: 8 guided modules, from a 4-bit ALU to an active bandpass filter, with Verilog export
- Wrote the core simulators from scratch: a nodal-analysis circuit solver driving the oscilloscope and Bode plotter, and a logic engine that flags floating inputs and shorts

TECHNICAL SKILLS

Languages Python, Dart, TypeScript, JavaScript, C/C++, Java, SQL, Verilog, Swift

AI/ML PyTorch Lightning, ONNX, TTS fine-tuning (Piper/VITS), speech-quality metrics, embedding search

Software Flutter, React, Firebase/Firestore, Cloud Functions, Drift/SQLite, Git, CI/CD (Codemagic)

Hardware Digital logic, breadboarding, oscilloscopes, ESP32, LTspice

Spoken languages Hebrew (native), English (fluent), Spanish (conversational)